Channel Estimation for Rydberg Receivers Using Learned Hankel Filtering

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***Abstract*—Rydberg receivers use excited atoms to sense radio signals. Magnitude-only observations do not directly reveal the wireless channel. Channel structure can help recover it, but fixed filtering may remove useful details. We study a small learned estimator that alternates measurement fitting with Hankel filtering. It learns from withheld observations, without true-channel targets. It improves on the original rank-search baseline, but a stronger classical shrinkage method has slightly lower average error and similar processing time. The learned method makes fewer severe errors in the tested noise mixture. Weak references and some few-path channels remain difficult. The result is a compact, label-free alternative with a measured accuracy–failure trade-off, not a generally superior channel estimator.**

***Index Terms*—Channel estimation, Hankel filtering, magnitude measurements, Rydberg receivers.**

I. Introduction

RYDBERG atoms respond to radio-frequency electric fields, offering a way to study new wireless receivers. Here, signals from several users combine with a known reference. The receiver measures their combined magnitude, not their full complex value. The channel describes changes in signal strength and phase on the way to the receiver. Model-based methods and learned methods such as URformer estimate it [1], [2].

A few propagation paths can create a repeated pattern across a regular array. Arranging channel samples in a Hankel matrix lets an estimator use this pattern. Strong filtering can remove real details; weak filtering can leave noise. Searching many filtering settings also takes time. We study a small model that learns how much filtering to apply. The true channel is never shown to it during training. Instead, some measurements are hidden from the estimator and used to check its predictions. This is called self-supervised learning.

Our contributions are: (i) a 188-parameter estimator that combines Rician measurement updates with smooth Hankel filtering; (ii) training this estimator on withheld measurements, without channel targets or channel-error checkpoint selection; and (iii) paired tests of accuracy, processing time and simpler versions, including fresh-test replication and input-removal tests. A controlled in-loop OptShrink comparison identifies a trade-off between average error and severe failures. We report losses alongside gains.

II. Related Work

Training on withheld measurements is established in image reconstruction [3] and self-supervised denoising is used for Rydberg sensor signals [4]. Huang et al. learn Hankel-factor updates from fully sampled targets [5]. Luo et al. combine self-supervision and structured low-rank reconstruction for complex MRI data [6]. DeepTensor fits network-based low-rank factors to an observed object [7]. Rician-likelihood training for magnitude data is also established [8]. Neither self-supervision, learned Hankel processing nor the Rician loss is new on its own.

We recover a complex multiuser channel from noisy magnitudes and a known reference, rather than restoring measured complex samples. A small controller sets filtering strength instead of generating low-rank factors. Our companion manuscript studies classical Hankel estimation and rank selection [9]; here we learn filtering without channel labels.

Structured alternating least squares [10], data-driven OptShrink [11], low-rank Hankel denoising (LRHD) [12] and robust Hankel completion [13] already provide other structural methods. Their original linear or direct matrix observations differ from our referenced magnitudes. Sparse amplitude-only beam training is also relevant [14], but uses adaptive sensing without our additive reference. We do not claim an optimal shrinker or superiority over these complete methods. Our stronger control puts OptShrink inside Rician EM iterations. This adapts data-driven shrinkage to the same observations; it is not a reproduction of the full LRHD algorithm for directly observed Hankel data.

III. Receiver and Channel Model

An array with N elements receives P known training symbols, or pilots, from each of K users. The observations are

$$Z = |GS + B + W|, \qquad W_{np} \sim \mathcal{CN}(0, v). \tag{1}$$

Here $G \in \mathbb{C}^{N\times K}$ is the channel, $S \in \mathbb{C}^{K\times P}$ contains the pilots, and $B \in \mathbb{C}^{N\times P}$ is the known reference. Magnitudes act entrywise. Independent noise has total complex variance v, or $v/2$ in each real component. The reference makes magnitudes depend on relative signal phases, but does not ensure accurate recovery in every case. Reference calibration and noise variance are known, except in stated input-mismatch checks. Receiver conversion gain is absorbed into the scaling. This effective model does not simulate atomic dynamics or optical readout.

For the half-wavelength uniform linear array,

$$g_{nk} = \sum_{\ell=1}^{L_k} \alpha_{\ell k} e^{-j\pi n \sin\theta_{\ell k}}, \quad n = 0, \ldots, N-1. \tag{2}$$

Independent gains follow $\mathcal{CN}(0, 1/L_k)$, giving each user unit average channel power. Angles are uniform on $[-\pi/2, \pi/2]$;

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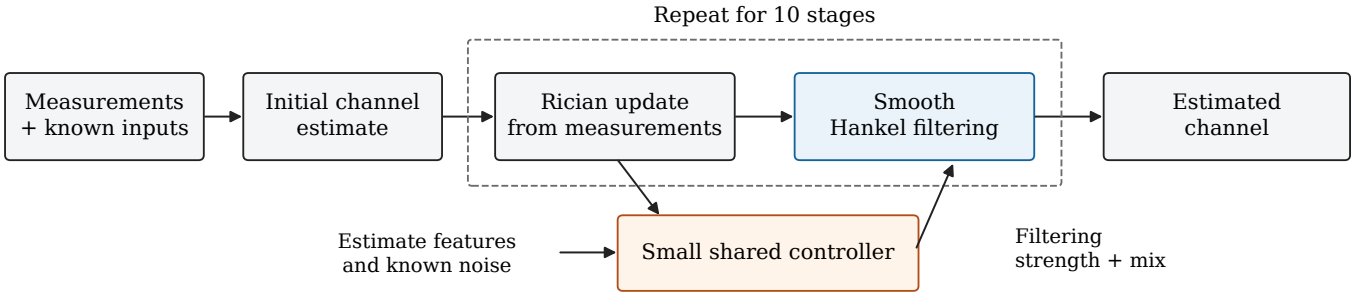


Fig. 1. Estimator overview. The controller chooses filtering strength and mixing. During training, nine withheld pilots score predictions but do not enter initialization or the ten stages.

pilot entries follow $\mathcal{CN}(0,1)$. The reference has a known spatial response and constant waveform. Define $\mathrm{SNR} = 10\log_{10}(K/v)$ and $\mathrm{RSR} = 10\log_{10}|\alpha_b|^2$, where α_b is reference amplitude. RSR uses one user's power: for three users, RSR 12 dB is about 7.23 dB relative to their combined power.

A. What the Measurements Can Determine

For one array element, write $\mu_p = g^T s_p + b_p$ and $x = [\mathrm{Re}\, g^T, \mathrm{Im}\, g^T]^T$. If $\mu_p \neq 0$, the real Jacobian of $|\mu_p|$ has row

$$A_p = \left[\mathrm{Re}\, c_p^T, -\mathrm{Im}\, c_p^T\right], \qquad c_p = \mu_p^* s_p/|\mu_p|. \tag{3}$$

Full column rank of A ensures local recovery; a small minimum singular value means poor stability. Thus $P \geq 2K$ is necessary for this condition, not sufficient. Full row rank of S alone is insufficient. With $B = 0$, channel rotation leaves magnitudes unchanged. A known reference can break this ambiguity, but its strength alone does not ensure recovery. This unrestricted-row condition is local, not a global or sparse-channel guarantee [15]. Proof and information matrix are in the supplement.

IV. Proposed Estimator

The estimator starts with a rough channel guess. It repeats two steps ten times: one pulls the guess toward the measurements, and the other smooths it using the array structure. A small network decides how much smoothing to apply at each step (Fig. 1).

A. Measurement Update

The magnitude of a complex signal with Gaussian noise follows a Rician distribution. For noiseless magnitude $a = |\mu|$, its density is

$$p(z \mid a, v) = \frac{2z}{v} e^{-(z^2+a^2)/v} I_0(2za/v), \tag{4}$$

where I_0 is a modified Bessel function. Let $M = GS + B$ and $\kappa = 2Z|M|/v$. Entrywise, form

$$Q = Z\frac{I_1(\kappa)}{I_0(\kappa)}\frac{M}{|M|}, \tag{5}$$

using the continuous limit at $M = 0$. The expectation-maximization (EM) update and learned step are

$$G_{\mathrm{EM}} = (Q - B)S^H(SS^H)^{-1}, \tag{6}$$

$$G_{\mathrm{dc}} = G + \gamma_t(G_{\mathrm{EM}} - G), \quad \gamma_t = 2\,\mathrm{sigmoid}(a_t). \tag{7}$$

A linear solve evaluates the inverse. Input pilots must have full row rank. Ten stages are a fixed computation budget, not a proved optimum.

B. Smooth Hankel Filtering

Each updated user channel forms a Hankel matrix $H = \mathcal{H}(g_{\mathrm{dc},k})$, with equal entries along each antidiagonal. At $N = 32$, it is 17×16. Singular values describe component strengths. Hard rank truncation deletes the smaller components. Our smooth rule is

$$\widehat{\sigma}_i = \frac{\sigma_i^3}{\sigma_i^2 + \tau^2}. \tag{8}$$

The multiplier $\sigma_i^2/(\sigma_i^2+\tau^2)$ is near one for large components and near zero for small ones. This can reduce noise but also bias genuine weak paths. Smoothness and a stable matrix solve motivate the choice; they do not make it an optimal denoiser.

Set $q = \lceil N/2 \rceil$, $s_H = \|H\|_F/\sqrt{q}$, and $\tau = s_H[0.001 + \mathrm{softplus}(u)]$. The equivalent operation is

$$\widehat{H} = H - \tau^2 H(H^H H + \tau^2 I)^{-1}, \tag{9}$$

$$g_k^{t+1} = g_{\mathrm{dc},k} + \eta_{t,k}(\mathcal{H}^\dagger \widehat{H}_k - g_{\mathrm{dc},k}), \tag{10}$$

where $\mathcal{H}^\dagger$ averages antidiagonals and $\eta_{t,k} = \mathrm{sigmoid}(w)$. The code normalizes by s_H before solving and guards zero scales. Gradients pass through the solve, without an identity-backward approximation. Neither control estimates path count. The filter does not increase the Hankel Frobenius norm, but this does not imply smaller channel error. The relaxed EM step decreases its quadratic surrogate for $0 \leq \gamma_t \leq 2$; the following filter need not decrease the likelihood. We make no convergence claim for the full estimator.

C. Controller Inputs

The network has eight inputs, 16 hidden units, two outputs and SiLU activation $x\,\mathrm{sigmoid}(x)$. Its 178 parameters and ten

step sizes give 188. For $p_i = \sigma_i / \sum_j \sigma_j$, its first three inputs are

$$e^{-\sum_i p_i \log p_i}/q, \qquad p_1, \qquad \sum_{i=q-\lceil q/2 \rceil+1}^{q} p_i. \tag{11}$$

They tell the network how spread out the components are, how much the largest contributes, and how much the smaller components contribute. Two more inputs measure disagreement with observations and expected noise relative to estimated channel power:

$$f_4 = \log\left(1 + \frac{\operatorname{mean}[(Z - |G_{\mathrm{dc}}S + B|)^2]}{v}\right), \tag{12}$$

$$f_{5,k} = \log\left(1 + \frac{v \operatorname{tr}[(SS^H)^{-1}]}{K\widehat{p}_k}\right), \quad \widehat{p}_k = \|g_{\mathrm{dc},k}\|^2/N. \tag{13}$$

Inputs $P/(2K)$, $\log q$ and $(t+1)/10$ give measurement count, size and stage ($t = 0, \ldots, 9$). Denominator guards avoid division by zero. True channels, path counts and family labels are not supplied. Weights stay fixed.

D. Training Without True Channels

Of 30 pilot columns, 21 enter the estimator and nine are withheld. Training scores predicted withheld magnitudes using the Rician loss

$$\ell(z, \mu, v) = \frac{(z - |\mu|)^2}{v} - \log I_{0e}(2z|\mu|/v). \tag{14}$$

Here I_{0e} is the scaled Bessel function. We omit an observation-only term and use small-field series at continuous limits. The loss fits Rician amplitude, not its mean. Splits change each epoch; separate validation cases with a fixed split select the checkpoint. Deployment uses all pilots, without extra measurements or weight training. This avoids test-driven selection; it does not make withheld likelihood equivalent to channel NMSE or remove dependence on the validation mixture.

V. Simulation Study

A. Setup and Comparators

Table I gives the settings. Half the training cases have three to seven paths per user; the rest use fixed counts of 2, 4, 8, 12 and 16. Validation weights 126 SNR, RSR and path-setting combinations equally. These condition labels set validation weights only; model selection uses withheld likelihood, not channel error.

The main study trains 18 models: two updates, three filtering choices and three seeds. Seeds change weights, batch order and pilot splits, but share training cases. Training took 3.67 hours with overlapping workers, excluding data preparation and preflight. Tests cover 32 conditions with 400 paired cases each. Two noise-input tests reuse cases, giving 12,000 distinct cases. All methods share channels, pilots and noise per case.

HS-GS, the companion study's classical Hankel estimator [9], fits ranks $1, \ldots, q$ on 70% of pilots, using 25 EM updates and four rank/Hankel projection sweeps per update. Squared magnitude residuals on the remaining pilots select a rank. It then refits all pilots for 100 updates. Rank search uses no extra observations and counts toward time. Its rank-selection loss differs from our Rician training loss. Other controls use EM, Rician or amplitude-gradient updates. Global filtering has 30 parameters and scales its threshold with s_H; no filtering has ten. Hard-Hankel Rician uses development-selected rank ten and one projection sweep per update. Constant-output controls replace adaptive outputs with validation averages without retraining.

The earlier URformer-style experiment uses a different protocol and is not a matched comparator here. Its scores are not pooled with these results.

Channel error is normalized mean squared error (NMSE):

$$\mathrm{NMSE}_{\mathrm{dB}} = 10 \log_{10} \frac{\sum_i \|\widehat{G}_i - G_i\|_F^2}{\sum_i \|G_i\|_F^2}. \tag{15}$$

Lower is better. Per-seed dB scores and the seven main SNR settings are averaged equally. Positive gain means lower error. A severe error has more energy than the true channel (per-case linear NMSE above one). Pointwise 95% intervals use 4,000 paired whole-case resamples, conditional on saved models, excluding training-dataset uncertainty. Classical stopping methods have a 1,000-update cap; reaching it need not mean convergence.

TABLE I
Main Simulation and Training Settings

Setting	Value
Array elements / users / pilots	32/3/30
Training / validation cases	16,384/2,048
SNR values (dB)	$-10, -5, 0, 5, 10, 15, 20$
Training RSR values (dB)	$5, 12, 20$
Training seeds	3, sharing training cases
Estimator stages / parameters	10/188
Training epochs / batch size	20/32
Optimizer / learning rate	Adam / 0.001
Input / withheld pilots	21/9
Training / evaluation precision	Single / double

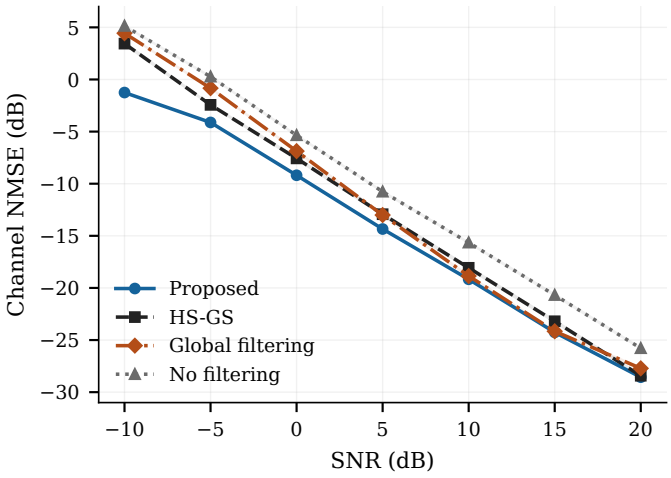


Fig. 2. Main SNR test: $N = 32$, $K = 3$, $P = 30$, RSR 12 dB, and three-to-seven-path channels. Lines average per-seed scores; bands are conditional paired-case 95% intervals.

TABLE II
ORIGINAL-STUDY SUMMARY; FULL COMPARATOR TABLE IN SUPPLEMENT

Method	NMSE (dB)	Time (ms)	Severe (%)
EM adaptive (primary)	-14.42	8.24	0.45
EM learned steps only	-10.41	2.98	22.11
EM global filtering	-12.43	5.61	19.19
GD adaptive (secondary)	-14.59	8.00	0.85
HS-GS, rank search	-12.74	525.62	14.25
EM, 100 updates	-10.28	17.01	23.57
Rician GD, hard Hankel	-11.06	47.39	20.93

NMSE and severe-error rates average seven primary SNR conditions equally. Time is the median at SNR 5 dB, RSR 12 dB. Learned scores average three seeds; classical cases are not counted three times.

B. Accuracy and Processing Time

HS-GS is the strongest original classical comparator on the declared average (Table II). The adaptive gradient-descent (GD) core has lower test NMSE than the primary EM core. We retain EM because validation likelihood selected it before test generation.

The EM model's average gain over HS-GS is 1.675 dB, with interval $[1.618, 1.730]$ dB and seed-specific gains of 1.533–1.787 dB (Fig. 2). At 20 dB, the gain is 0.139 dB with interval $[-0.017, 0.294]$ dB, which does not establish a clear improvement.

Severe errors fall from 14.25% for HS-GS to 0.45% for the proposed method.

Median time is 8.24 ms versus 525.62 ms for HS-GS, about 64 times faster (Table II). Timing includes initialization and rank search, using one Apple M2 CPU thread, double precision, batch size one, 100 cases and three repeats at SNR 5 dB and RSR 12 dB. These measurements depend on the implementation: PyTorch for learned methods and NumPy/SciPy for classical methods. Offline training is separate. With $m = N - q + 1$, one adaptive stage costs $O(NKP + NK^2 + K^3 + Kmq^2 + Kq^3)$: the spectral features still require an SVD. HS-GS repeats projections across ranks and updates. A small weight count does not imply low computation.

C. Operating Limits

The proposed method loses 1.617 dB at RSR 0 dB, 1.104 dB with two paths, and 0.203 dB with three paths (Fig. 3). Good average performance does not ensure good performance with a weak reference or a simple channel.

Transfer tests at SNR 5 dB and RSR 12 dB give gains for variance inputs wrong by ± 3 dB and for clustered or ten-path alternatives with matched energy. One-factor dimension tests cover $N \in \{8, 16, 32, 64\}$, $K \in \{2, 3, 4\}$ and $P \in \{10, 20, 30\}$. At $N = 8$ the gain interval includes zero. A new 32-case-per-cell transfer check extends to $P = 6 = 2K$: at SNR 5 dB, learned NMSE is -2.197 dB, compared with -14.332 dB at $P = 30$. Full local Jacobian rank does not prevent poor finite-noise accuracy. The supplement defines all alternatives and reports dimension results. Path counts overlap training.

TABLE III
FRESH STRONGER-BASELINE COMPARISON, SEVEN-SNR AVERAGE

Method	NMSE (dB)	Severe (%)
Rician-ranked HS-GS	-12.890	13.06
Final-only OptShrink	-14.374	3.79
In-loop OptShrink	-14.462	5.02
Learned ten-stage model	-14.319	0.89

D. Further Comparisons

A separate 15-model follow-up uses two fresh test sets and six adjusted primary comparisons. Two independent training datasets, one seed each, give gains of 1.939 and 1.837 dB over HS-GS. Replication is still limited. A matched supervised recipe is worse on the declared average but better in some cells; its gap changes with selection and averaging. This does not establish general superiority of self-supervision. Repeated filtering gains about 0.70 dB over a retrained final-only model at extra cost. Mixing stays near one: the controller mainly changes shrinkage strength, not a reliable on/off decision. All follow-up results remain in the supplement.

E. Which Inputs Help?

Input-removal controls train on 4,096 shared cases for 12 epochs with three seeds; two fresh tests total 2,304 worlds. Full inputs improve average error over spectrum-only, noise/residual-only and size/stage-only controllers. However, they lose 0.39–0.40 dB against noise/residual-only at SNR 5 dB, failing the declared 0.25 dB protection rule, and 0.82–0.93 dB on two-path cases. Both input groups help the average, not every channel. Size/stage inputs, 188 stored parameters, and the physical update/filter remain in all controls; unused input weights are masked.

F. A Stronger In-Loop Shrinkage Comparison

A separate controlled study uses 224 development cases and 1,152 fresh test cases: 128 at each of seven SNRs and two stress conditions. The learned models remain frozen. OptShrink rank, mixing and iteration count are chosen on withheld Rician likelihood, separately by known noise level, before test generation. The same grid includes a final-only control and no filtering. No true channel enters selection; all methods use the same test measurements.

In-loop OptShrink wins the declared average by 0.143 dB, with conditional paired interval $[0.086, 0.198]$ dB (Table III). The learned model has fewer severe errors; these occur at SNR -10 dB in this sample. Mean per-case linear NMSE favors the learned model (0.197 versus 0.215), showing that the ranking depends on the declared metric. This does not reverse the primary average-dB result. At SNR 5 dB, mean end-to-end times are 5.67 and 6.20 ms for learned and in-loop methods, including noise-based setting lookup. Thus the large original speed ratio does not apply to this stronger baseline. At a weak reference, longer OptShrink runs also beat the learned model; transferring a stopping rule can fail.

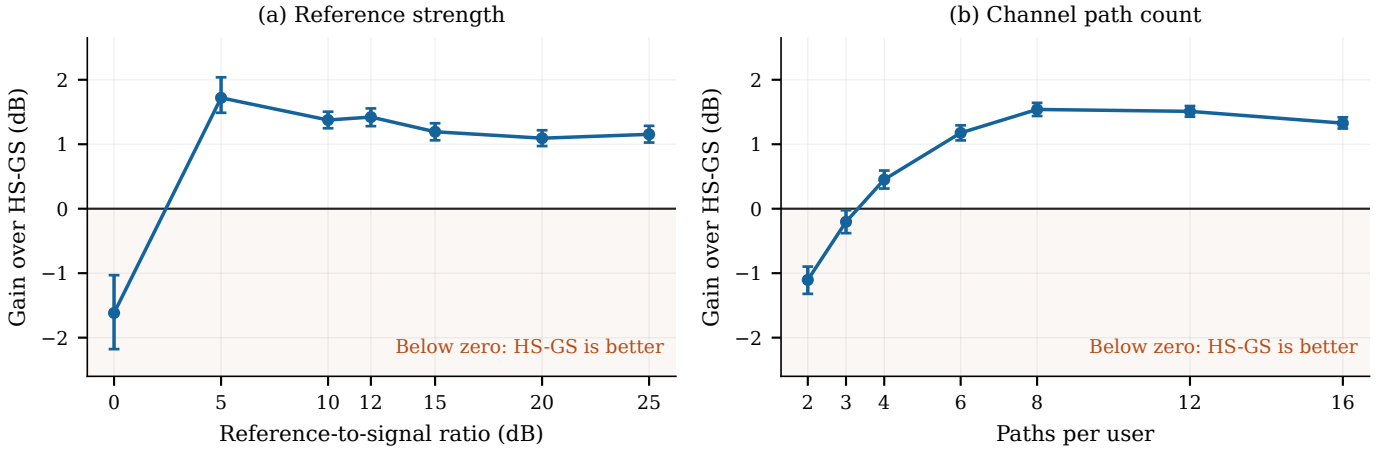


Fig. 3. Limits at SNR 5 dB: (a) reference strength with three-to-seven-path channels; (b) fixed path counts at RSR 12 dB. Positive gain favors the proposed estimator. Bars are conditional pointwise paired-case 95% intervals. Path counts overlap training.

VI. Discussion and Conclusion

An additional study trains 25 models on 4,096-case datasets for 12 epochs and tests 3,584 fresh worlds. Five stages lose 1.977 dB against ten; twenty gain 0.181 dB but take 15.60 rather than 8.04 ms. Width 32 and two experts gain only 0.065 and 0.023 dB. A strong-to-weak-reference curriculum gains 0.206 dB on average but loses 2.353 dB in its worst cell. No variant improves every condition. The supplement records selected/final scores and independent-data variation; these exploratory tests do not replace the original model.

An exploratory static noise/reference-amplitude correction helps some input errors but hurts some weak-reference cases; it cannot determine unknown reference phase. A nearly noiseless stress test at SNR 60 dB, far outside training, also exposes filtering bias: at RSR 12 dB the learned estimate reaches only -6.647 dB NMSE, versus -65.715 dB for EM100. Disabling filtering with the same learned steps gives -44.020 dB in a post-hoc diagnostic. Low-noise consistency is not established.

A small model can learn useful Hankel filtering without true-channel targets. Its practical setting is repeated estimation near the simulated training population, where offline training is reused. It is better than the original rank-search baseline, but not the stronger in-loop control on average dB error. Its lower severe-error frequency is a separate, conditional finding, not evidence of general superiority.

All results are simulations. Neither global recovery, full-model convergence nor superiority over published URformer is established. Case-resampling intervals exclude training-data uncertainty. The new curriculum includes weak references but does not remove all failures. Atomic dynamics, optical readout, unknown reference drift and two-dimensional arrays remain untested. The supplement retains full comparisons and numerical safeguards.

References

[1] B. Xu, J. Zhang, Z. Chen, B. Cheng, Z. Liu, Y.-C. Wu, and B. Ai, "Channel estimation for Rydberg atomic receivers," *IEEE Wireless Communications Letters*, vol. 14, no. 9, pp. 2957–2961, 2025, author manuscript: arXiv:2503.08985.

[2] J. Xiao, J. Wang, M. Zeng, H. Xu, X. Li, and A. Nallanathan, "Channel estimation for Rydberg atomic quantum receivers: Unrolled phase retrieval from holographic snapshots," *IEEE Signal Processing Letters*, vol. 33, pp. 1696–1700, 2026.

[3] B. Yaman, S. A. H. Hosseini, S. Moeller, J. Ellermann, K. Ugurbil, and M. Akcakaya, "Self-supervised learning of physics-guided reconstruction neural networks without fully sampled reference data," *Magnetic Resonance in Medicine*, vol. 84, no. 6, pp. 3172–3191, 2020.

[4] Z. Liu, Q. Ren, W. Yang, Y. Tong, H. Wang, Y. Zhang, R. Zhi, J. Xie, M. Jing, H. Zhang, L. Xiao, S. Jia, K. Tang, and L. Zhang, "Self-supervised learning with noisy dataset for Rydberg microwave sensors denoising," arXiv:2601.01924, 2026.

[5] Y. Huang, J. Zhao, Z. Wang, V. Orekhov, D. Guo, and X. Qu, "Exponential signal reconstruction with deep Hankel matrix factorization," arXiv:2007.06246v4, 2021, accepted author manuscript, IEEE Transactions on Neural Networks and Learning Systems.

[6] C. Luo, H. Wang, T. Xie, Q. Jin, G. Chen, Z.-X. Cui, and D. Liang, "Matrix completion-informed deep unfolded equilibrium models for self-supervised k-space interpolation in MRI," arXiv:2309.13571, 2023.

[7] V. Saragadam, R. Balestriero, A. Veeraraghavan, and R. G. Baraniuk, "DeepTensor: Low-rank tensor decomposition with deep network priors," arXiv:2204.03145, 2022.

[8] C. S. Parker, A. Schroder, S. C. Epstein, J. Cole, D. C. Alexander, and H. Zhang, "Rician likelihood loss for quantitative MRI using self-supervised deep learning," arXiv:2307.07072, 2023.

[9] A. Haatim, "Hankel-structured channel estimation for magnitude-only Rydberg atomic receivers: When does structure help?" 2026, unpublished companion manuscript.

[10] D. Zachariah, M. Sundin, M. Jansson, and S. Chatterjee, "Alternating least-squares for low-rank matrix reconstruction," *IEEE Signal Processing Letters*, vol. 19, no. 4, pp. 231–234, 2012.

[11] R. R. Nadakuditi, "OptShrink: An algorithm for improved low-rank signal matrix denoising by optimal, data-driven singular value shrinkage," *IEEE Transactions on Information Theory*, vol. 60, no. 5, pp. 3002–3018, 2014.

[12] M. Yin and R. S. Smith, "On low-rank Hankel matrix denoising," *IFAC-PapersOnLine*, vol. 54, no. 7, pp. 198–203, 2021.

[13] H. Cai, J.-F. Cai, and J. You, "Structured gradient descent for fast robust low-rank Hankel matrix completion," *SIAM Journal on Scientific Computing*, vol. 45, no. 3, pp. A1172–A1198, 2023.

[14] Z. Wang, S. Tsai, Y. Hu, and R. Zhang, "GP bandit-assisted two-stage sparse phase retrieval for amplitude-only near-field beam training," arXiv:2603.07477, 2026.

[15] B. Gao, Q. Sun, Y. Wang, and Z. Xu, "Phase retrieval from the magnitudes of affine linear measurements," *Advances in Applied Mathematics*, vol. 93, pp. 121–141, 2018.